

Structured Computer Architecture
Chapter 2.9: Memory Circuits
First-Order Systems (1-OS)
An Order-Based Approach

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Based on the textbook

September 14, 2025

Outline

Introduction to First-Order Systems

Elementary Latches

Clocked Latches and Timing

Flip-Flops and Edge-Triggered Circuits

Timing Analysis and Constraints

Random Access Memory (RAM)

Synchronous Memory Systems

Register Files

Advanced Memory Concepts

Chapter Summary

Learning Objectives

By the end of this chapter, you will be able to:

- ▶ Understand the concept of first-order systems (1-OS) and feedback loops
- ▶ Analyze the behavior of latches and flip-flops
- ▶ Design clocked memory circuits and understand timing constraints
- ▶ Explain the operation of RAM and register files
- ▶ Distinguish between synchronous and asynchronous memory systems
- ▶ Understand the trade-offs in memory circuit design

From 0-OS to 1-OS: Adding Memory

- ▶ **0-OS Limitation:** Output depends only on current inputs
 - ▶ No memory of previous states
 - ▶ Cannot store information
 - ▶ Purely reactive behavior
- ▶ **1-OS Innovation:** Introduction of the first feedback loop
 - ▶ Output fed back to input
 - ▶ Creates memory capability
 - ▶ Enables state storage
 - ▶ Partial autonomy achieved
- ▶ **Key Principle:**
 - ▶ Memory function obtained through reverse connection
 - ▶ Circuit maintains state according to command
 - ▶ Foundation for all sequential logic

The Feedback Loop Concept

▶ **Feedback Mechanism:**

- ▶ Output signal returns to influence input
- ▶ Creates dependency on previous state
- ▶ Enables information storage

▶ **State Autonomy:**

- ▶ State can be maintained without continuous input
- ▶ Temporary input can have permanent effect
- ▶ System "remembers" past events

▶ **Applications in Computing:**

- ▶ Register storage in processors
- ▶ Memory cells in RAM
- ▶ State machines in control units
- ▶ Pipeline registers

The Simplest Memory Element

Latch Function: Store temporary signal transitions

- ▶ Capture transition from 1 to 0 (reset-only latch)
- ▶ Capture transition from 0 to 1 (set-only latch)
- ▶ Maintain state indefinitely

Basic Principle:

- ▶ Feedback loop from output to input
- ▶ Signal propagation through circuit
- ▶ State becomes self-sustaining

Limitation:

- ▶ Simple latches cannot "forget"
- ▶ Need combination for full functionality
- ▶ Asymmetric command signals

Elementary Latches

